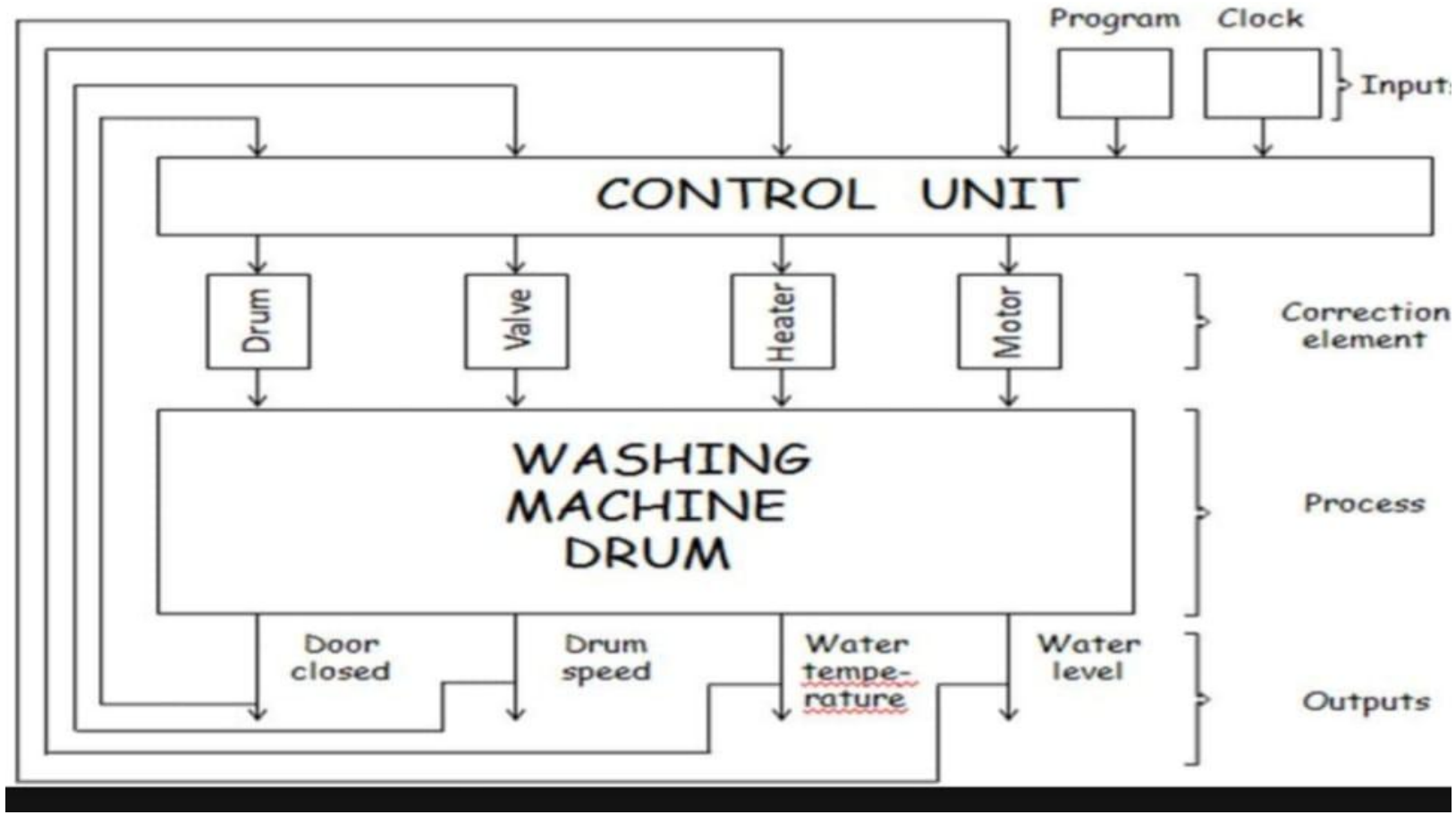
# AUTOMATIC WASHING MACHINE

Principle:

> Automatic washing machine is based on “Centrifugal force.”
(The apparent outward force on mass when it is rotated.)

> Closed loop control system is employed.

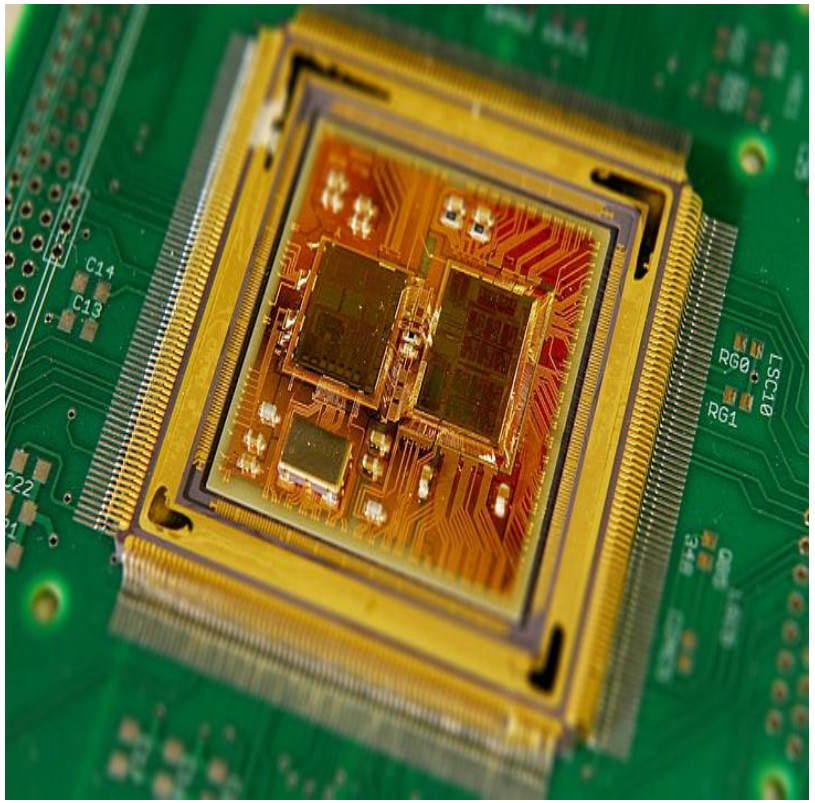
# WHAT ARE THE PARTS OF WASHING MACHINE?



# Microprocessor:

- The chip contains a CPU, Some memory and a program or software, which the washing machine manufactures made.

  The Chip has a number of Input and Output pins on it... Once the software is started ( buy selecting different pre set programs using the selector switch or buttons) The chip monitors the states (on or off) the inputs, E.g Is drum filled with water? Yes = 1 Now =0 The program will move on to turn drum motor on, when the state is 1. It does this by switching the motor turn output pin on 1=on 0=off.... The PIC can make use of special functions, like timer delays, turn drum on for x amount of time... once x time is counted down the program moves on.



# HOW IT WORKS?

- With modern washing machines, the controller is a microprocessor and the program is not supplied by the mechanical arrangement of cams but by a software program.
- The microprocessor controlled washing machine can be considered an example of a mechatronics approach in that a mechanical system has become integrated with electronic controls.

- For the pre-wash cycle an electrically operated valve is opened when a current is supplied and switched off when it ceases.
- This valve allows cold water into the drum for a period of time determined by the profile of the cam or the output from the microprocessor used to operate its switch.
- A sensor is used to give a signal when the water level has reached the preset level and give an output from the microprocessor which is used to switch off the current to the valve.
- In the case of a cam-controlled valve, the sensor actuates a switch which closes the valve admitting water to the washing machine drum.
- When this event is completed, the microprocessor, or the rotation of the cams, initiates a pump to empty the drum.

- For the main wash cycle, the microprocessor gives an output which starts when the pre-wash part of the program is completed; in the case of the cam-operated system the cam has a profile such that it starts in operation when the pre-wash cycle is completed.
- It switches a current into a circuit to open a valve to allow cold water into the drum. This level is sensed and the water shut off when the required level is reached.
- The microprocessor or cams then supply a current to activate a switch which applies a larger current to an electric heater to heat the water. A temperature sensor is used to switch off the current when the water temperature reaches the preset value.
- The microprocessor or cams then switch on the drum motor to rotate the drum. This will continue for the time determined by the microprocessor or cam profile before switching off. Then the microprocessor or a cam switches on the current to a discharge pump to empty the water from the drum.

- The rinse part of the operation is now switched as a sequence of signals to open valves which allow cold water into the machine, switch it off, operate the motor to rotate the drum, operate a pump to empty the water from the drum, and repeat this sequence a number of times.
- The final part of the operation is when the microprocessor or a cam switches on just the motor, at a higher speed than for the rinsing, to spin the clothes.